Esteban Saldarriaga-Marín

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Profile and interests

I am an MSc in Mathematics student at Universidad de los Andes. My main interests are Number Theory and Arithmetic Geometry, and I work under Professor Jerson Caro. I held a position of Visiting Scholar at Purdue University in which I worked on elliptic curves and modularity under guidance of Professor Daniel Le and Professor Guillermo Mantilla.

Education

MSc in Mathematics — Aug 2025 -
Universidad de los Andes — Bogota, Colombia

BS in Mathematics — Aug 2019 - Aug 2024
Universidad Nacional de Colombia — Medellin, Colombia

Relevant coursework

Universidad de los Andes

- An introduction to Iwasawa Theory (graduate) . . . . . Jan 2026 - May 2026
- Hilbert's Tenth Problem and extensions to number fields (graduate) . . . . Jan 2026 - May 2026

Universidad Nacional de Colombia

- Class Field Theory (graduate)* . . . . . Nov 2024 - Mar 2025
- Analytic Number Theory (graduate)* . . . . . Nov 2024 - Mar 2025
- Introduction to Advanced Algebraic Topology . . . . . Aug 2023 - Dec 2023
- Introduction to Group Representation Theory . . . . . Aug 2023 - Dec 2023
- Introduction to the Theory of Elliptic Curves (graduate) . . . . . Aug 2023 - Dec 2023
- Algebraic Topology I . . . . . Feb 2023 - Jun 2023
- Algebraic Number Theory (graduate) . . . . . Feb 2023 - Jun 2023

*Just attended as was no longer enrolled in the University

Research experience

Visiting Undergraduate Student (REU) — Feb 2024 - Aug 2024
Purdue University — West Lafayette, IN

Modularity Theorem has many equivalent versions. The a_p-version strongly uses the striking relation between Hecke operators modulo p and the Frobenius element. This relation is known as the Eichler-Shimura relation. With Prof. Le we deeply studied all the relevant material to state and reach a proof of this relation. Currently, with assistance from Prof. Mantilla, I am studying other approaches to this result via other equivalent formulations of Hecke operators.

Teaching experience

Teaching Assistant *Universidad de los Andes, Mathematics Department*
Teaching and advisory sessions to students, preparation and revision of teaching material, exams grading

- Multivariable Calculus
- Differential Calculus
- Integral Calculus with Differential Equations

Teaching Assistant *AGRA V*
Advisory sessions to students and preparation and revision of teaching material

- Métodos explícitos de Chabauty: un punto de vista computacional a los puntos racionales

Undergraduate Teaching Assistant *Universidad Nacional de Colombia, Mathematics Department*
Advisory sessions to students, preparation and revision of teaching material, exams grading

- Multivariable Calculus
- Precalculus

Awards and honors

- *Undergraduate tuition waiver* by Universidad Nacional de Colombia 2020

Conferences and workshops attended

- *AGRA V - A school on Arithmetic, Groups and Analysis*, Universidad del Cauca. July and August of 2026.
- *GARLAT - Latin American School in Arithmetic Geometry*, Universidad Católica de Chile. July of 2026.
- *Preliminary Arizona Winter School on Local Fields*, The Southwest Center for Arithmetic Geometry. September to November of 2024.
- *Geometric, Algebraic and Topological Methods in Quantum Field Theory*, CIMPA. Villa de Leyva, Colombia, July 2023.
- *Second Colombian Conference on Applied and Industrial Mathematics*, Colombian Mathematical Society. Medellin, Colombia, June 2022.

Volunteering

- *Food Finders food bank*, volunteering at Fresh Market pantry. Lafayette, Indiana, June 2024.

Skills

- **Programming languages**
  - Python (beginner)
  - Mathematica (beginner)
  - Matlab (beginner)
  - LaTeX (advanced)
- **Languages**
  - English (TOEFL iBT – 95)
  - Spanish (native)